Algebra 1: Unit 9, Lesson 4

1. The revenue, in billions of dollars, for a company in the year 2003 was \$2.5 billion. One year later, in 2004, the revenue had risen to \$3.2 billion. In 2007, the revenue climbed to \$4.1 billion, before falling to \$3.2 billion in 2010.

   The revenue, r, in billions of dollars, for the company is a quadratic function of the number of years since 2003, x.

   What is the vertex of the function?

2. An object is launched from the ground and reaches a maximum height of 56.6 meters after 3.4 seconds. The object returns to the ground after 6.8 seconds.

   Jayden is asked to write a quadratic function to model the height of the object, H, as a function of the time, t, in seconds since it was launched.

   Jayden uses the vertex of $(3.4, 56.6)$ to write the equation, $H(t) = a(x - 3.4)^2 + 56.6$. Then, Jayden uses the point where the object hits the ground of $(6.8,\ 0)$ to find the value of a. What is the value of a?

3. The water stream from a water fountain forms a parabola. The water stream begins flowing 1 inch (in.) from the edge of the drain at a height of 1.3 in . When the water is 4 in. from the right edge of the drain, the water reaches its highest point at a height of 4.9 in. The water returns to a height of 1.3 in. when it is 7 in. from the right edge.

   Use the information to complete the first step in determining the function that models the water stream where x is the distance, in inches, from the right edge of the drain, and y is the height of the water stream, in inches.

   $$\boxed{} = a\left(7 - \boxed{}\right)^2 + \boxed{}$$